Šimun Batovanja Electronic Engineer & Software Developer

Electronics engineer with hands-on experience in power electronics, embedded systems, PCB design, and software development across the automotive and telecommunications industries.

CONTACT

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Portfolio: https://s-batov.github.io/Portfolio/

EXPERIENCE

Electronics team member/lead | Nov. 2021 – Oct. 2026

FESB Racing, Formula Student team

- Led the electronics team, coordinating design, testing, and integration of vehicle electronic systems with the software and mechanical teams
- Designed, manufactured and tested custom electronic hardware for vehicle systems
- Developed and assembled the complete wiring harness for the race car
- Configured the traction inverter for vehicle operation, collaborating with the powertrain and control teams

Electronics/Software Intern | Mar. 2026 – Aug. 2026

Audi Formula Racing GmbH

- Designed, assembled and debugged prototype PCBs for a Formula 1 power unit
- Developed a 48V power management device, multiple adapter/breakout boxes, revised multichannel power supply
- Manufactured and verified mixed-signal subcircuit prototypes using in-house fabrication and assembly equipment
- Conducted end-of-line testing for ECUs and helped operate the Electronics Lab and inventory system

Hardware/Software Engineer | Sep. 2024 – Jan. 2026

Axialis j.d.o.o.

- Developed PCB schematics and layouts for inverter control circuits
- Performed SPICE simulations to estimate losses and verify design performance
- Designed and manufactured battery pack wiring

Software developer | Aug. 2022 – Sep. 2024

Ericsson Nikola Tesla d.d.

- Developed and implemented software features in the Sync and Inter-connect modules of radio equipment
- Ran JCAT tests and monitored CI pipelines, improving testing efficiency and system reliability
- Coordinated with R&D teams to integrate and validate new radio functionalities

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EDUCATION

Master of Engineering (Electronics)

Faculty of Electrical Engineering, Mechanical Engineering and Naval Architecture

2023 – 2026

Bachelor of Science (Electrical Engineering & IT)

Faculty of Electrical Engineering, Mechanical Engineering and Naval Architecture

2019 – 2023

CERTIFICATES

HV handling according to DGUV 209-093 stage 3E

Rimac Technology - Evalus

Jun. 2024

SKILLS

Software & Tools

- CAD/PCB: Altium Designer, Fusion 360
- Simulation: Altium SPICE simulator, LTspice
- Embedded: STM32CubeIDE
- Version control: Git

Hardware / Technical Skills

- Power electronics design
- DSP and microcontroller programming
- Wiring harness design and manufacturing
- PCB prototyping, testing and debugging with lab equipment

Broader Engineering Skills

- System integration and testing
- Inverter control and configuration
- CI loops and automated testing